

Program:

```
#include <stdio.h>
#include <stdlib.h>
#define MAX 100
// Maximum size of the queue
// Queue structure definition
struct Queue {
    int arr[MAX];
    int front;
    int rear;
};
// Function prototypes
void initializeQueue(struct Queue *queue);
int isFull(struct Queue *queue);
int isEmpty(struct Queue *queue);
void enqueue(struct Queue *queue, int element);
int dequeue(struct Queue *queue);
void displayQueue(struct Queue *queue);
int main() {
    struct Queue queue;
    int choice, element;
    initializeQueue(&queue);
    while (1) {
        printf("\nQueue Operations Menu:\n");
        printf("1. Enqueue\n");
        printf("2. Dequeue\n");
        printf("3. Display\n");
        printf("4. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);
        switch (choice) {
            case 1:
                printf("Enter element to enqueue: ");
                scanf("%d", &element);
                enqueue(&queue, element);
                break;
            case 2:
                printf("Enter element to dequeue: ");
                scanf("%d", &element);
                element = dequeue(&queue);
                if (element != -1)
                    printf("Dequeued element: %d\n", element);
                break;
            case 3:
                displayQueue(&queue);
                break;
            case 4:
                return 0;
        }
    }
}
```

```
break;
case 4:
exit(0);
default:
printf("Invalid choice! Please enter a valid option.\n");
}
}
return 0;
}
```

Output:

Queue Operations Menu:

1. Enqueue
2. Dequeue
3. Display
4. Exit

Enter your choice: 1

Enter element to enqueue: 10

Element enqueued: 10

Queue Operations Menu:

1. Enqueue
2. Dequeue
3. Display
4. Exit

Enter your choice: 1

Enter element to enqueue: 20

Element enqueued: 20

Queue Operations Menu:

1. Enqueue
2. Dequeue
3. Display
4. Exit

Enter your choice: 1

Enter element to enqueue: 30

Element enqueued: 30

Queue Operations Menu:

1. Enqueue
2. Dequeue
3. Display
4. Exit

Enter your choice: 3

Queue elements: 10 20 30

Queue Operations Menu:

1. Enqueue
2. Dequeue

3. Display

4. Exit

Enter your choice: 2

Dequeued element: 10

Queue Operations Menu:

1. Enqueue

2. Dequeue

3. Display

4.Exit

Enter your choice: 3

Queue elements: 20 30

Queue Operations Menu:

1. Enqueue

2. Dequeue

3. Display

4. Exit .